Palo Alto Networks Cybersecurity Academy – Factory Resetting a PA220 Firewall And Accessing WEB GUI

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Purpose

The purpose of this lab is to give a introduction to the world of cyber-security by documenting one of the most basic function of a networking device: Factory reset. This lab teaches users how to manage, and factory rest a Palo Alto 220 firewall. It also teaches users how to get into the firewall web Gui for further management.

Background

Palo Alto networks is a global cybersecurity leader. Founded in 2005 by Nir Suk, it provides a platform for network, cloud, and security operations. They are a member of the S&P 500. They focus mainly on the business market, creating scalable security solutions for many of the largest companies worldwide. The company's core product, the Next-generation Firewall (Introduced in 2007), revolutionized network security by integration application, user, and content awareness.

The Palo Alto Networks PA-220 next-generation firewall, contrary to their core product, is designed for small organizations or branch offices and includes the following main features: active/passive and active/active high availability (HA), passive cooling (no fans) to reduce noise and power consumption, eight Ethernet ports, and dual power adapters for power redundancy. The PA-220 firewall enables you to secure your organization through advanced visibility and control of applications, users, and content.   Marketed as a NGFW, Next-generation firewall. The PA220 uses machine learning to identify attacks instead of relying on simple signature check like traditional firewalls. This technology allows the PA220 to identify undocumented threats and brand-new exploits without intervention from Palo Alto networks themselves. The PA220 also prevents threats by filtering URLs and securing against DNS-based attacks.

Since the PA220 is a hardware firewall, meaning it's a physical device, it is a more tangible device as compared to a virtual software firewall. By using this legacy form factor, the PA220 can provide a higher degr   ee of reliability as the physical device can be troubleshooted by the end user. With a software firewall, security is managed remotely by a 3rd party company, meaning that any issues with the outside company could leave software firewalls users vulnerable for hours or even days. Hardware firewalls put this control in the hands for the users, allowing security to be implemented immediately by an experience technician.

One major feature of software-based firewalls is that they can be managed remotely without a physical or local network connection to the firewall. The PA220, despite its form factor, also supports remote management through Palo Alto's Panorama software. Panorama supports a wide range of Palo Alto firewalls and allows them all to be remotely monitored and managed from a single dashboard.

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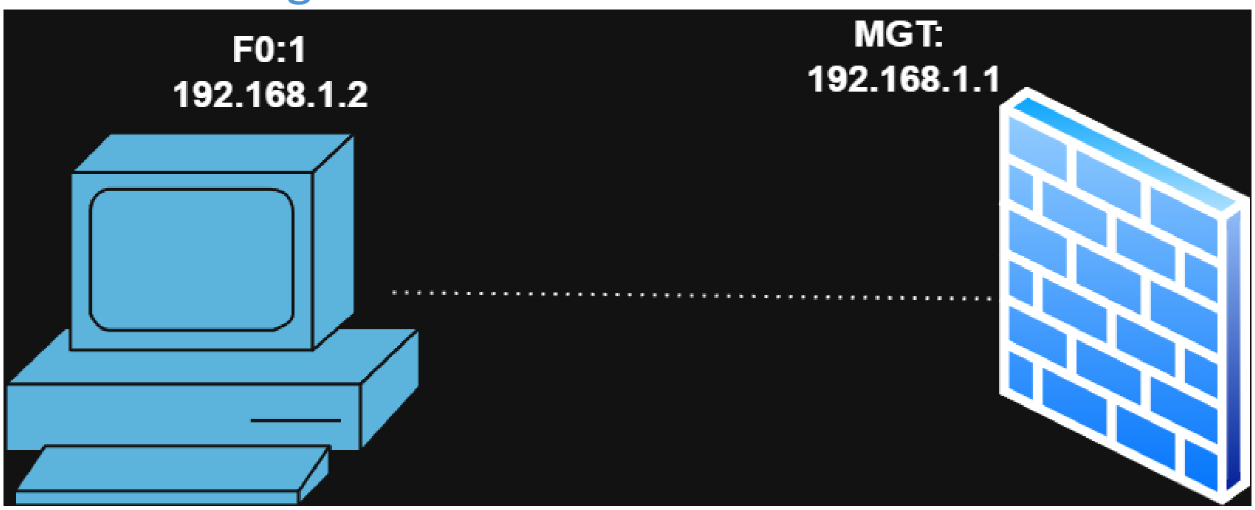
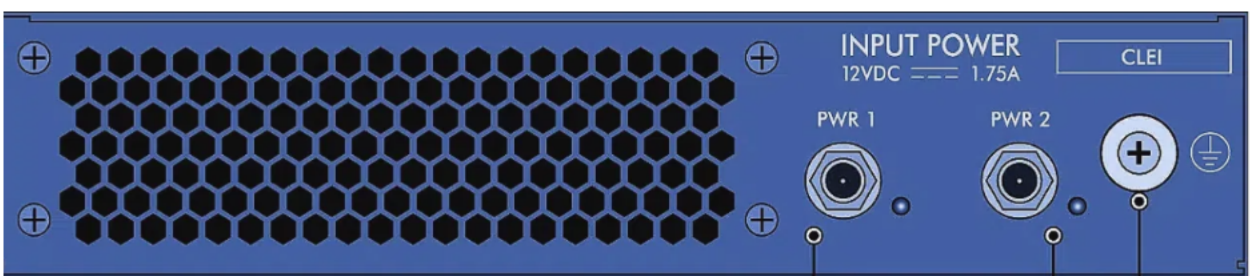
The Palo-alto 220 runs a piece of software called PAN-OS. In this lab, our firewall is running PAN-OS 9.1, which is an older version of the software that has reached end-of life. The software is accessible through multiple interfaces, including through a console connection and through an HTTP connection on a local network. By default, PAN-OS is protected through an account called admin, with a default password set to admin.

Network Diagram

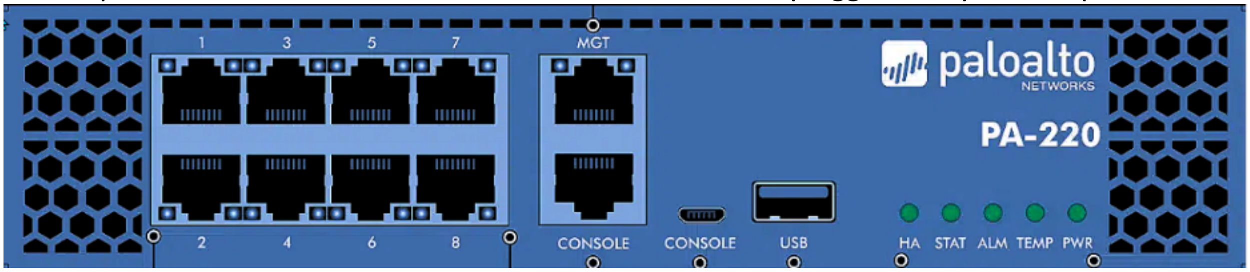


Lab Procedure

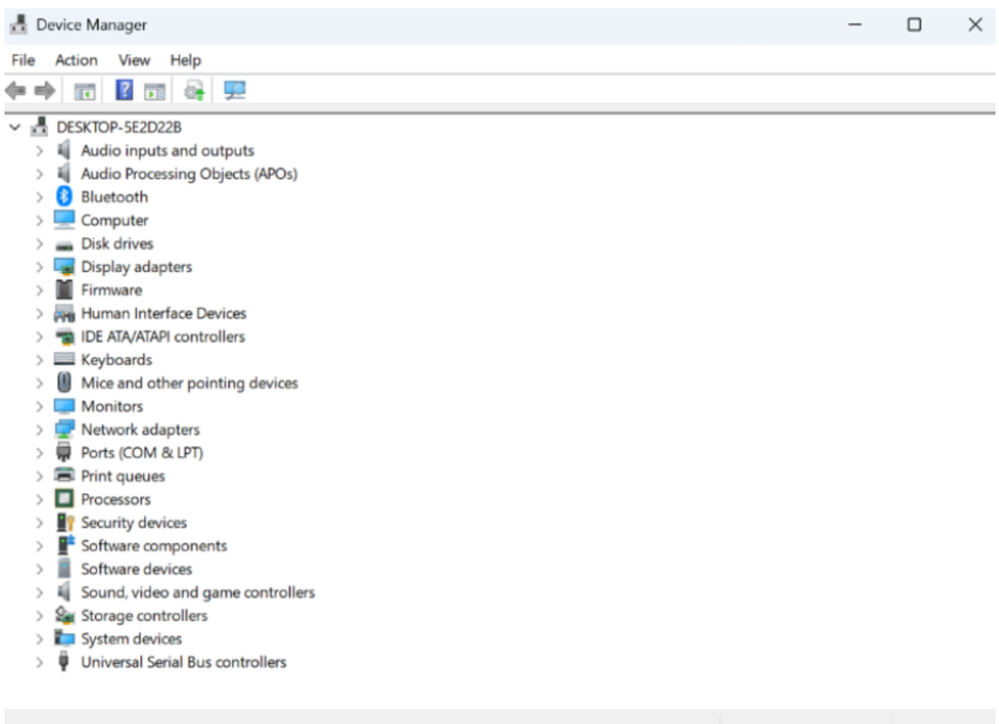
1. Plug in your power cord into either the PWR 1 or 2 port and the other side of the power cord into your computer.



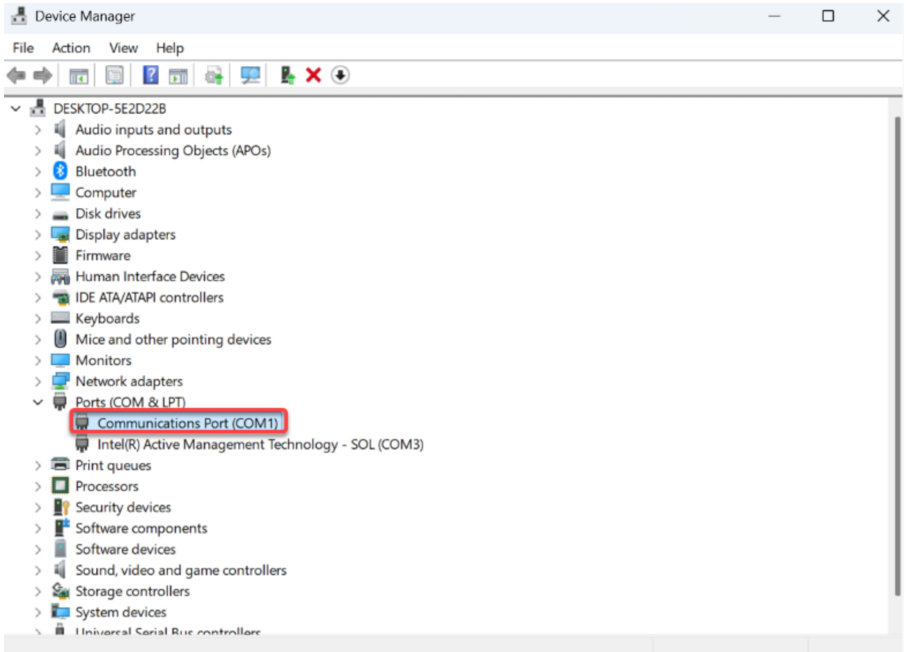
2. Insert your Console Cable into the port labeled Console and the Ethernet Cable into the port labeled MGT with the other side of these cables plugged into your computer.



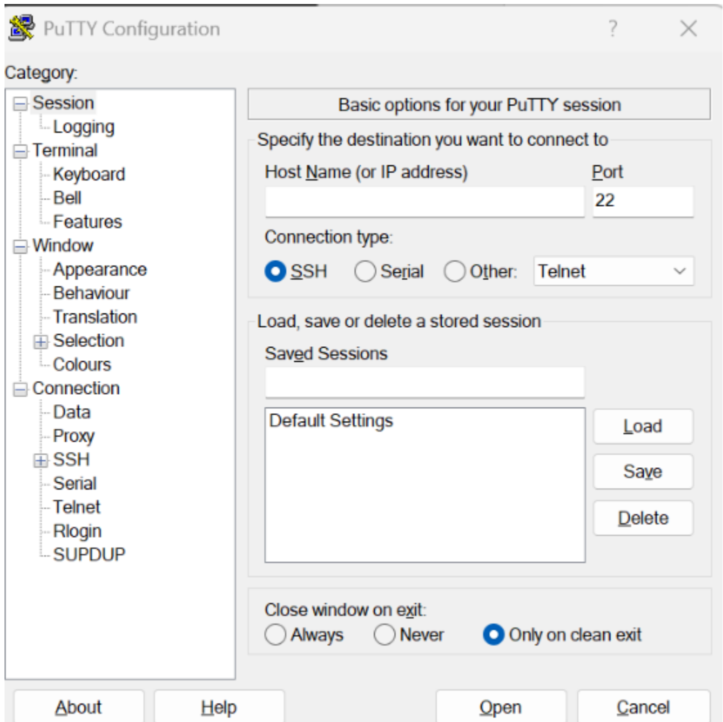
3. Click on the windows button found on the taskbar on the bottom of the screen and type in Device Manager before pressing the enter button. You should be directed to a place screen as shown above.



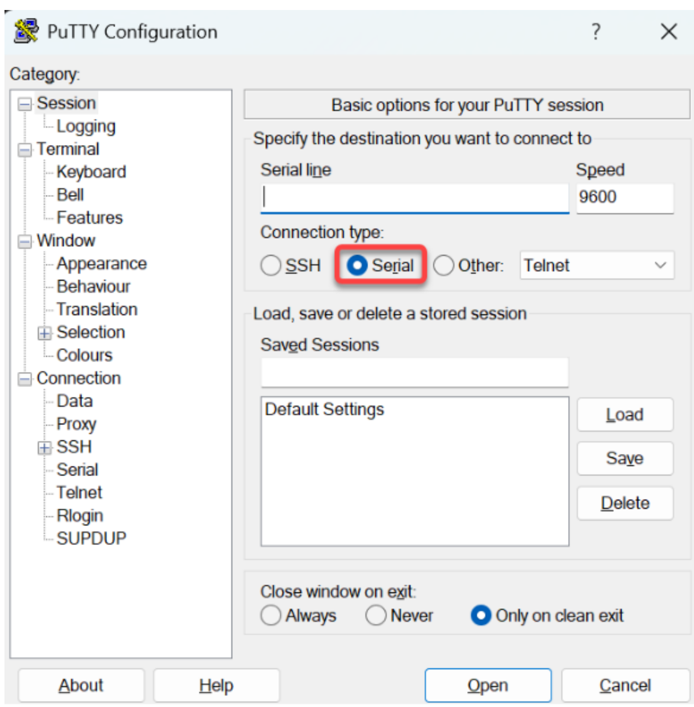
4. Click on the arrow that is to the side of the Ports (COM & LPT) section which shows 2 more sections. Note the specific COM port that is found as highlighted above and in this situation that would be COM1.



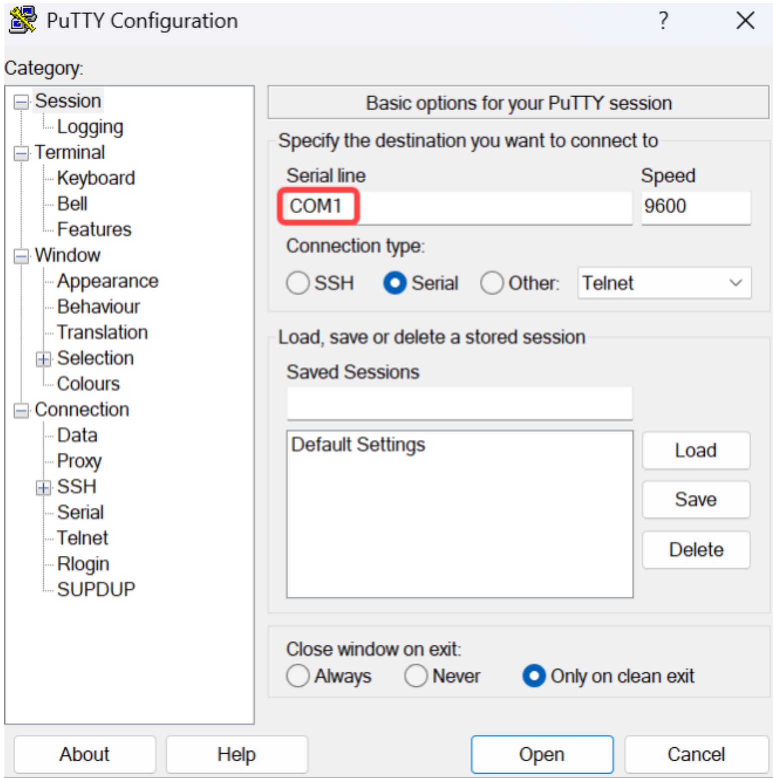
5. Type "PuTTY" into the windows icon that is on the taskbar and press the enter button. This should bring you to a window similar to the image above.



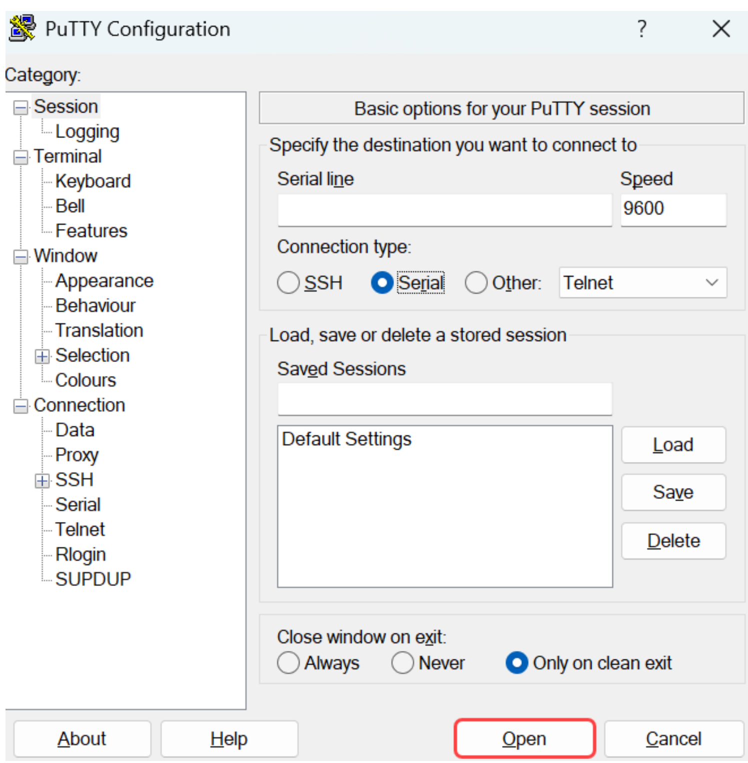
6. Press the Serial circle just as higlighted above.



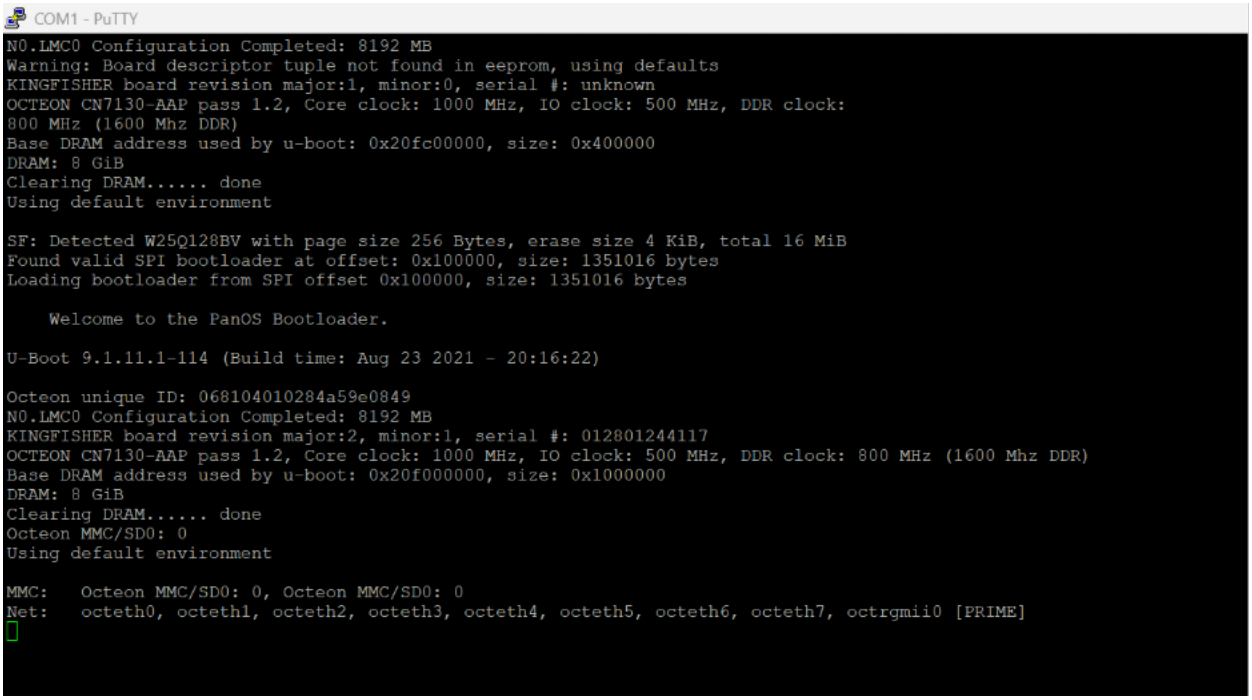
7. Enter your COM port as noted before in a pervious step as highlighted above.



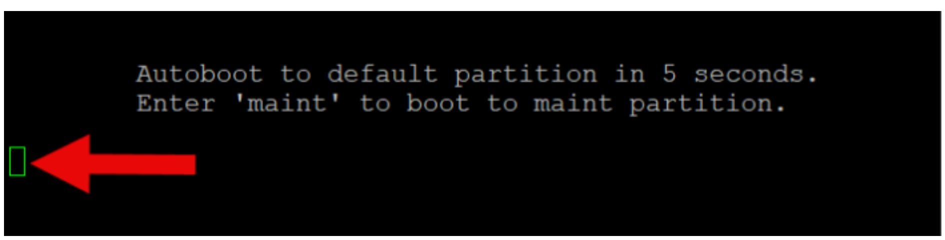
8. Select the Open button just as highlighted above.



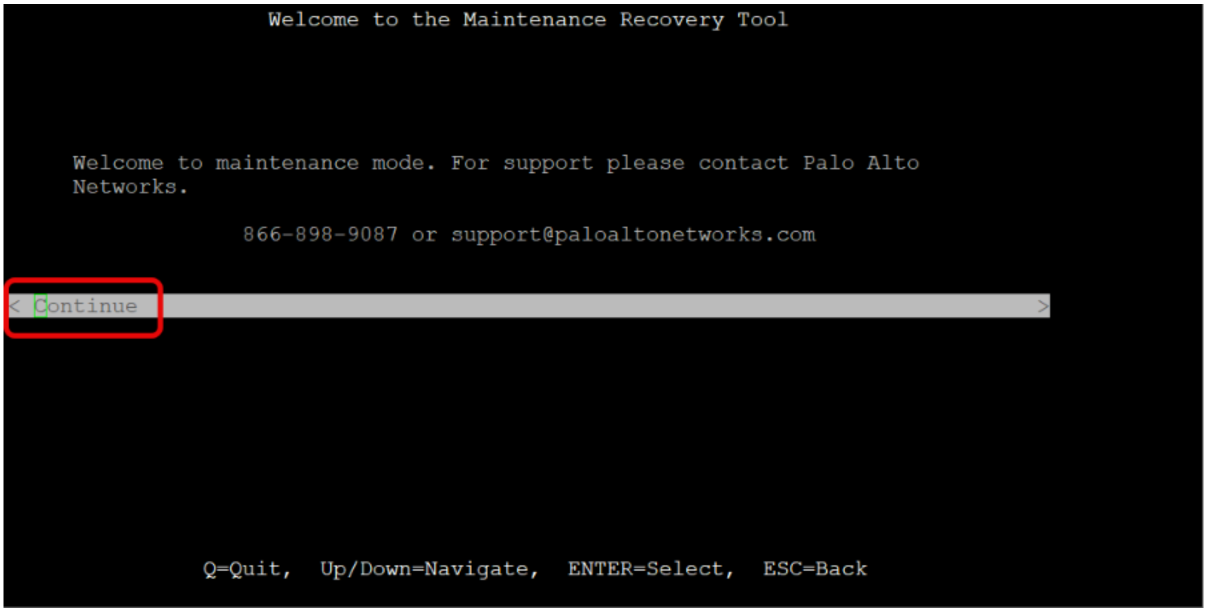
9. You should then be brought to a screen similar to the image above



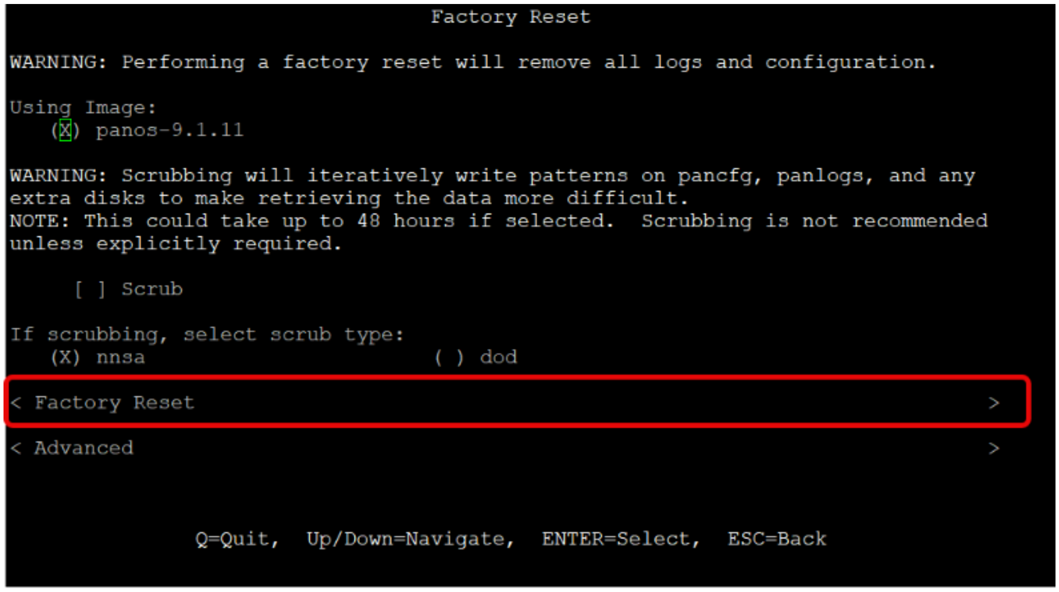
10. Wait at the screen until you get a message in text exactly like the image above and enter the wors “maint” within 5 seconds in the terminal as highlighted above.



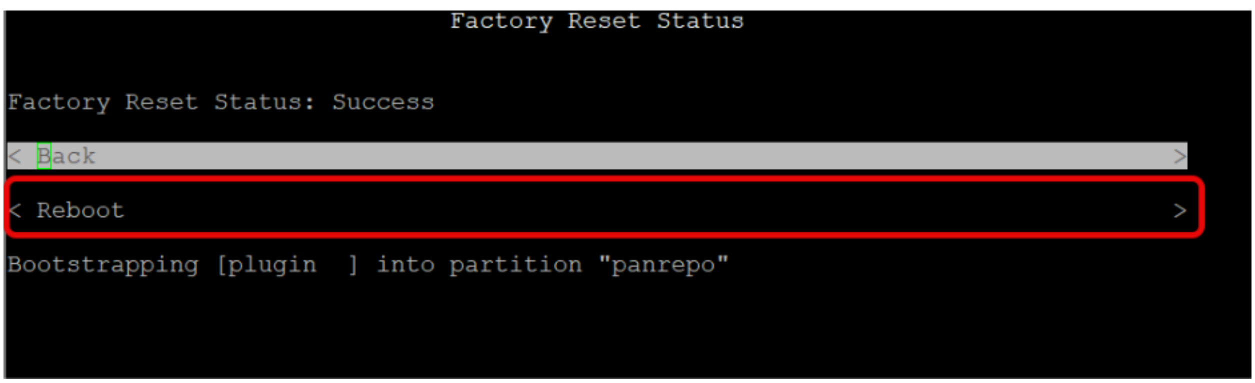
11. After waiting, you should then be brought to a screen as shown in the image above and then press the enter button as highlighted above.



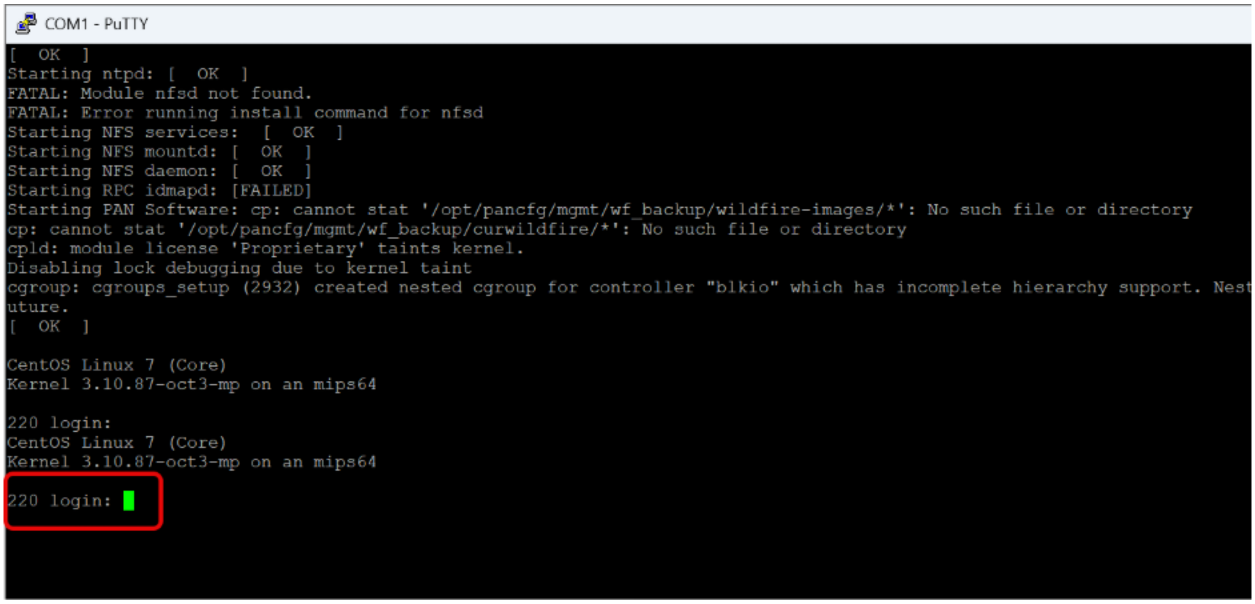
12. Press the down button until you get to the Factory Reset section as highlighted above and press the Enter button.



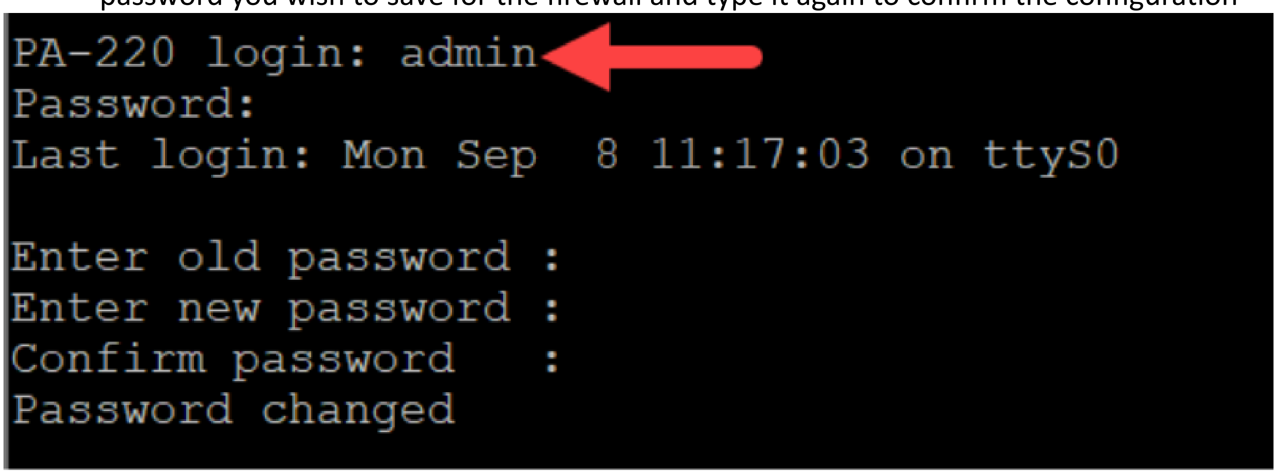
13. After waiting for the firewall to complete the Factory Reset and being brough to this screen, click the down arrow and press Enter on the Reboot section as highlighted above.



14. Then you should be brought to a screen like the image above and the terminal should say 220 login.

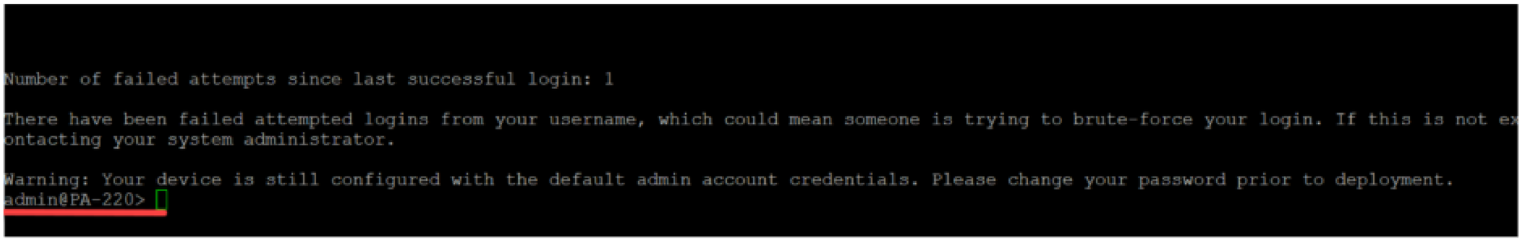


15. Type in a login username like admin as arrow indicates in the image above that you wish to save as the new configuration. Next type in a corresponding password. Type any string of words for the old password section. Then press the enter button, type in the password you wish to save for the firewall and type it again to confirm the configuration

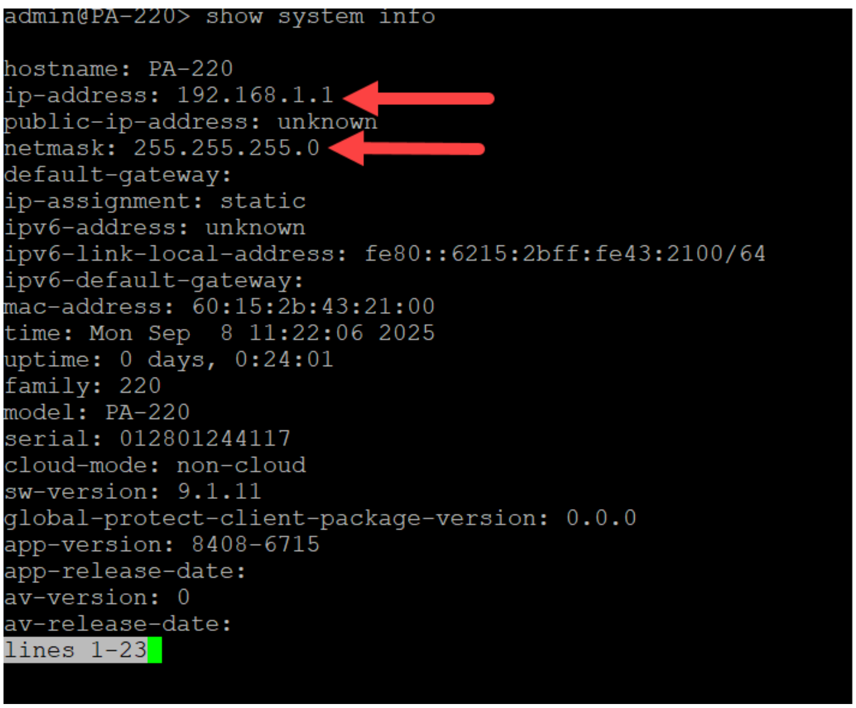


16.
The firewall will then display text that indicates that the password is still not configured, and the previous password is still the account credential. However, you want to ignore

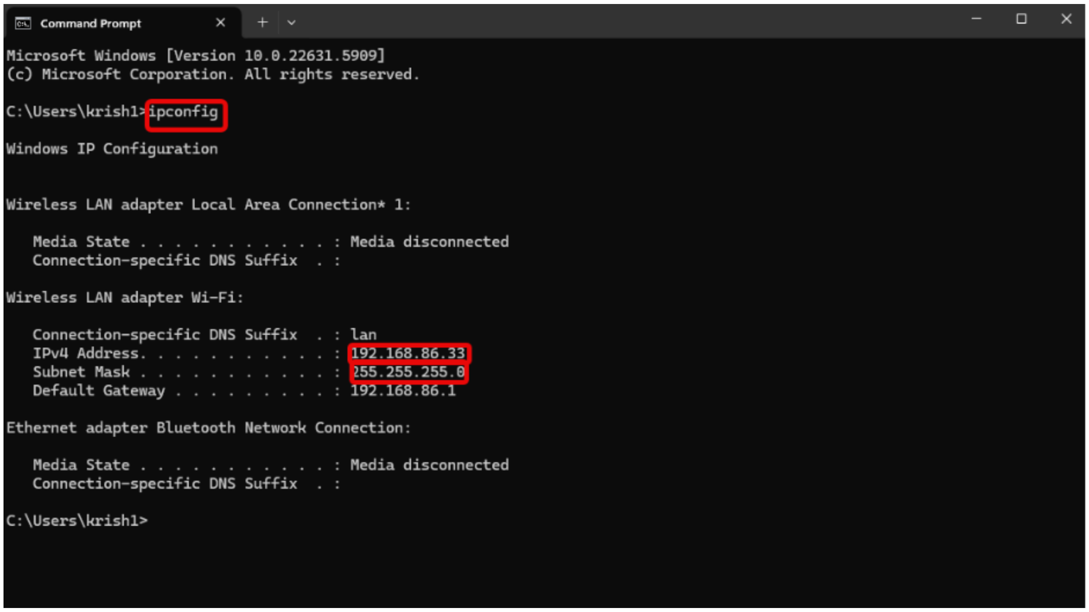
this message and check that the terminal displays your username followed by @PA-220> to see if the configuration is temporarily saved. If so continue to the next steps, and if not unplug the firewall power cord and Factory Reset the firewall once again until you complete this step.



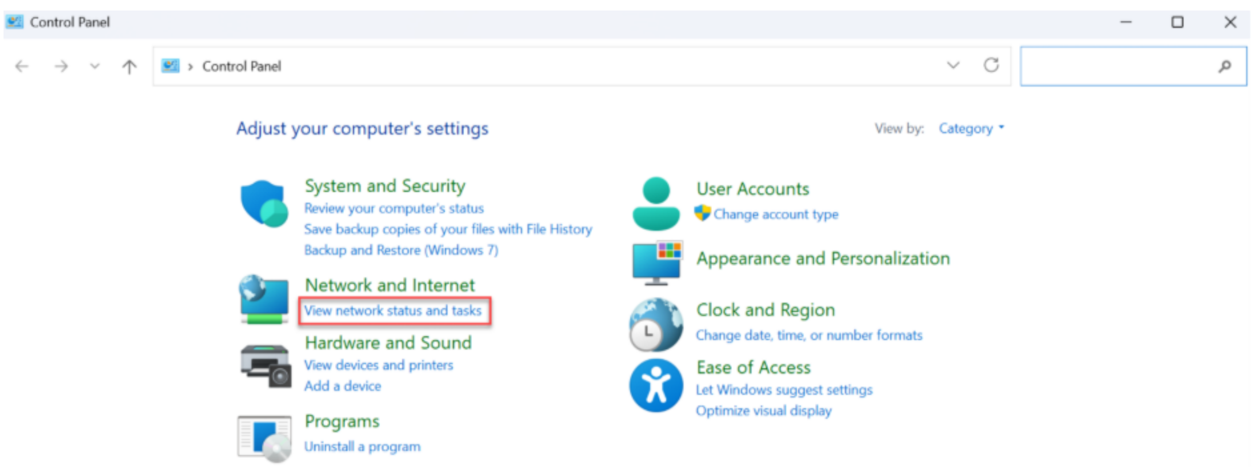
17. Then type in the command “show system info” as shown in the above image. Be sure to take note of the Ip address and netmask that should be in a similar format as shown by the arrows in the image above.



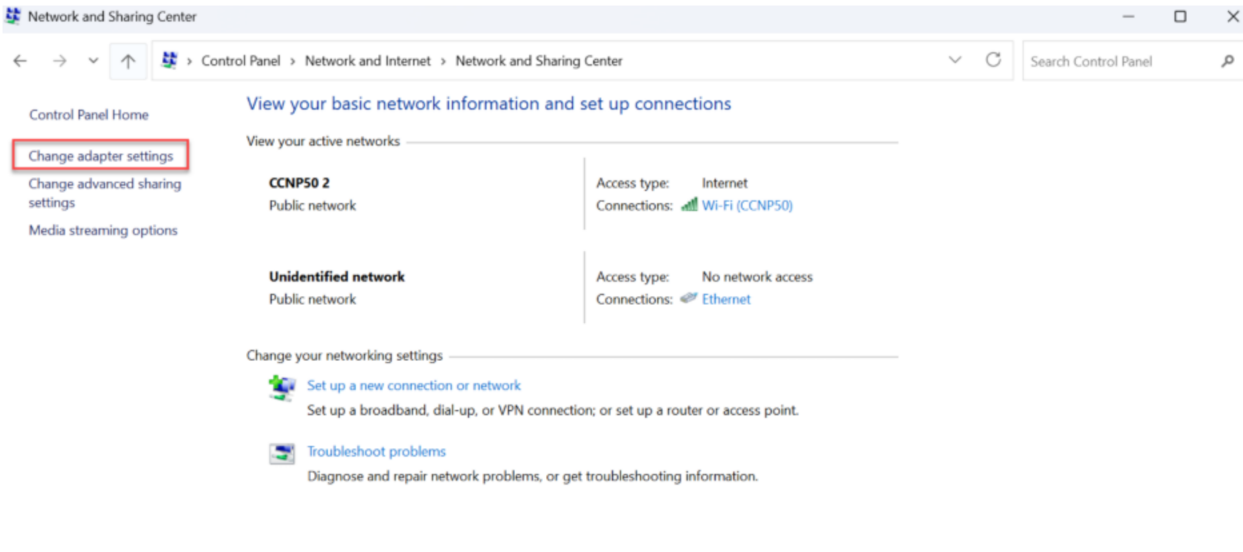
18. Type in Command prompt after clicking the Windows Icon and type the ipconfig command as highlighted in the picture above. Be sure to note down the IPv4 Address and the Subnet Mask as highlighted above.



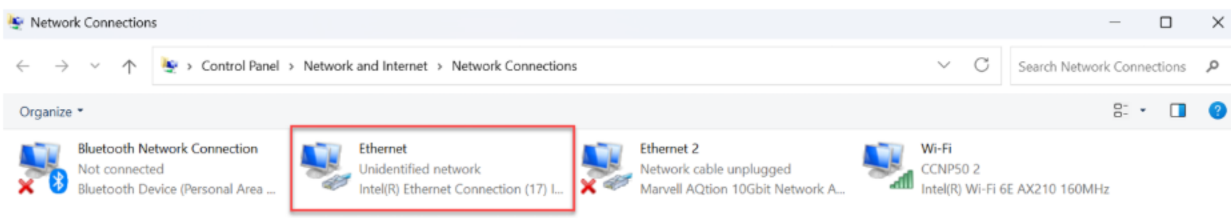
19. Type in Control Panel after clicking on the Windows Icon in the taskbar at the bottom of the screen and after being brought to a window similar to the image above be sure to click the View network and status and tasks section as highlighted above.



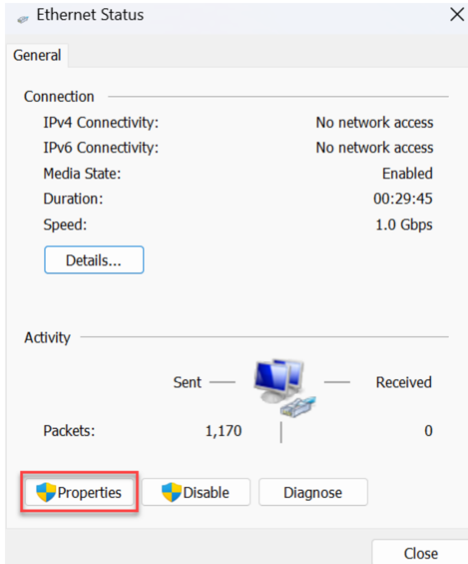
20. After being brought to a windows similar to the image above, click the Change adapter settings as highlighted above.



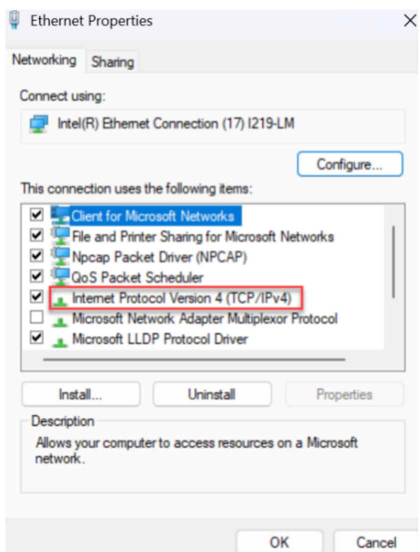
21. Click twice on the Ethernet Section as highlighted above.



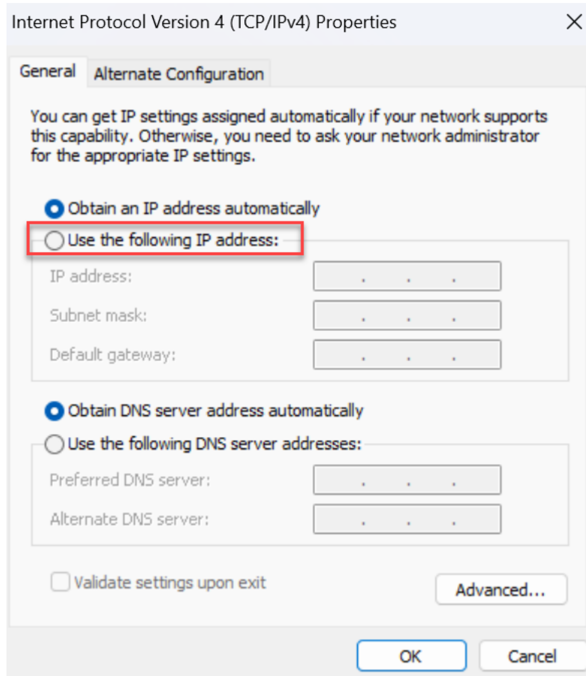
22. After being brought to an Ethernet Status page similar to the image above, click twice on the Properties button as highlighted above.



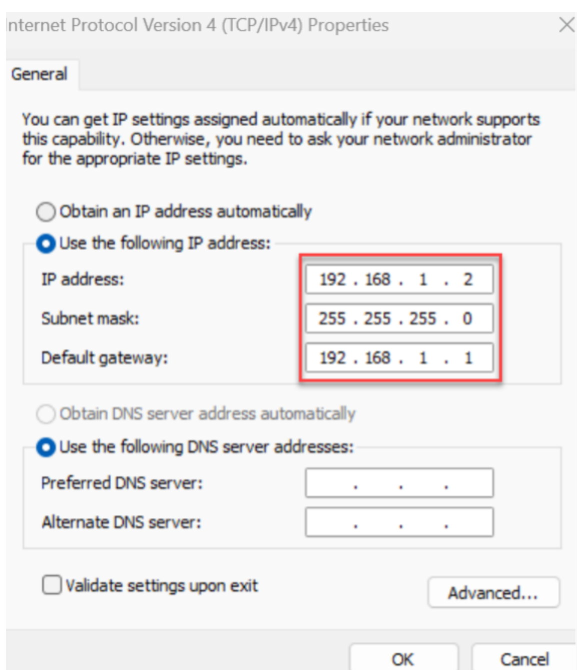
23. After opening the following window as shown in the image above, be sure to click twice on the Internet Protocol Version 4 (TCP/IPv4) as highlighted above.



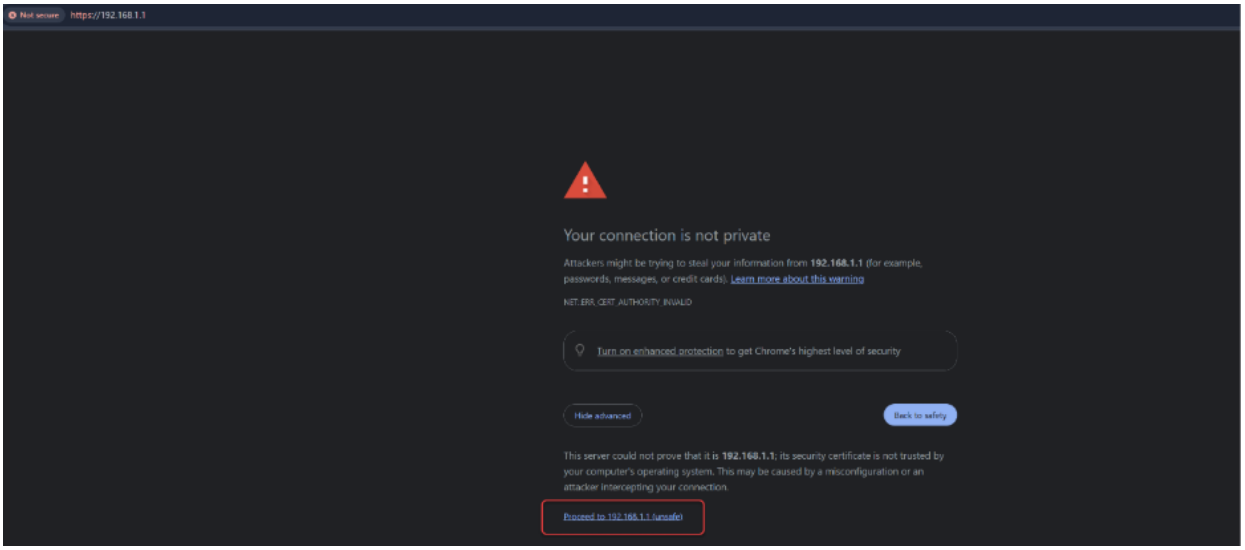
24. At this point you should see a window exactly like the image above and be sure to click the circle next to the Use the following IP address as highlighted above.



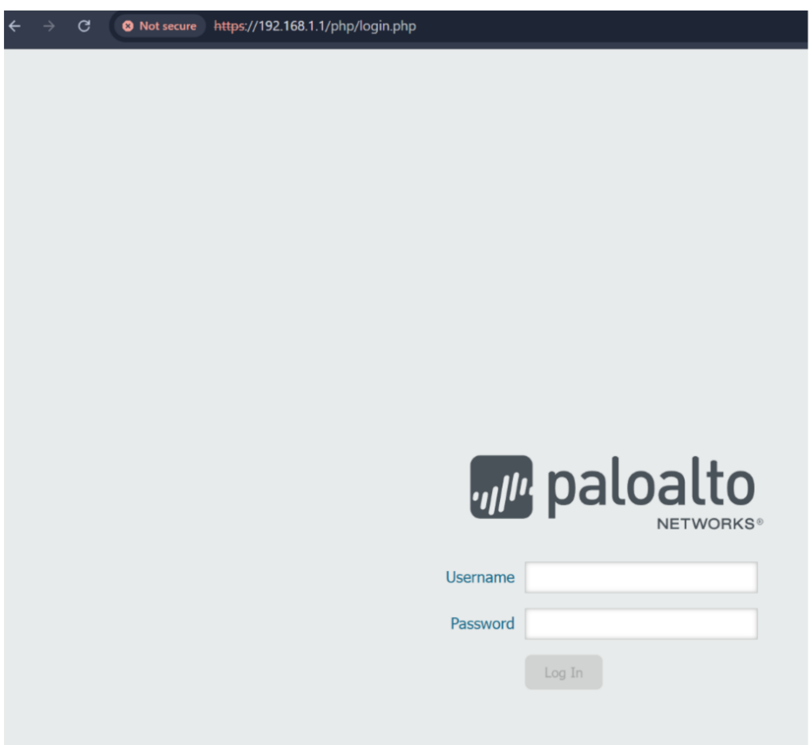
25. Type in the IP address and Subnet mask noted down from the Command prompt from a previous. For the Default Gateway be sure to write down the numbers that you noted down from the terminal of the Firewall.



26. Go to your browser of choice and type https://_______ with the underline being the IP Address of the Firewall that you noted down from the terminal of the Firewall from a previous step. Then click the show advanced button as highlighted above and then the Proceed to ____(unsafe) where the underline is the IP Address of the Firewall noted from a previous step.



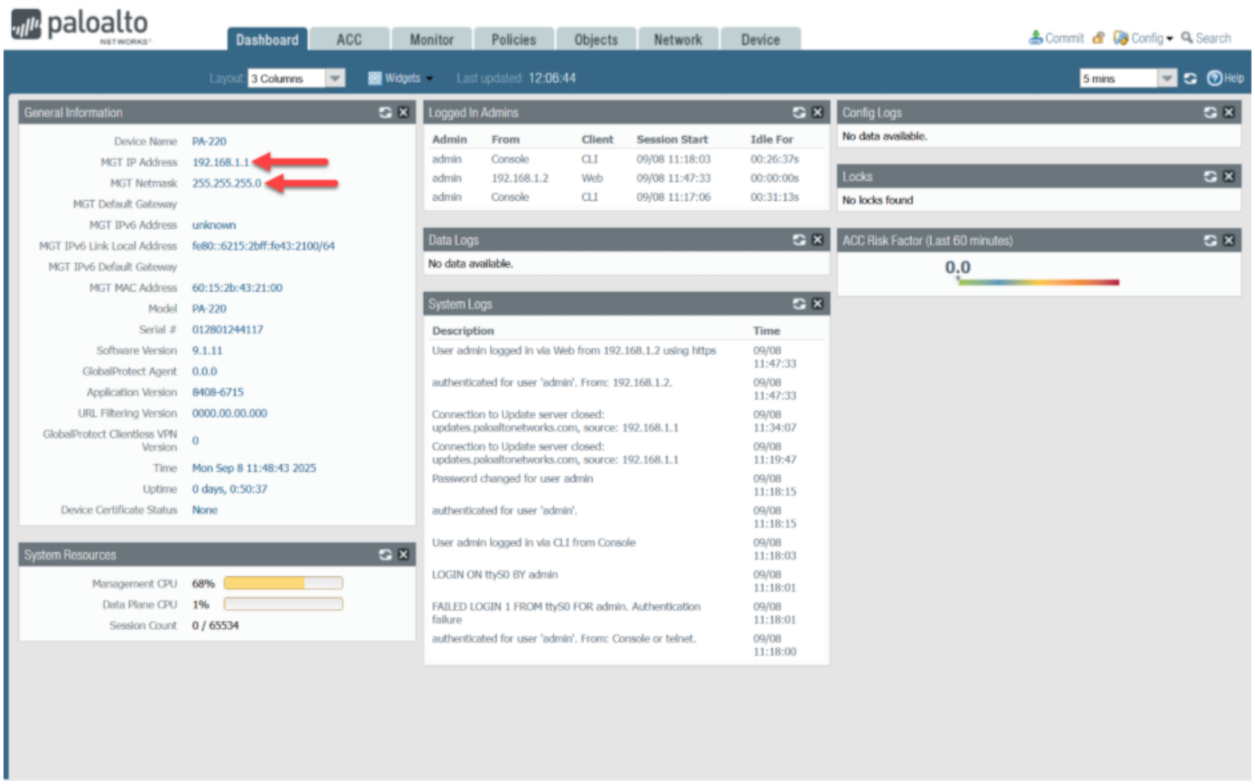
27. The window should then transition into a webpage similar to the image shown above.



28. Enter in your username and password that you had temporarily set up from a previous step in the boxes as highlighted above. Then press the Log In button.

29. After being redirected to the next page a window should appear showing the current PAN-OS Software to which you should press the Close button as highlighted above.

30. In the Web GUI as shown above you will be able to see various details such as the Firewall's IP Address, Netmask as indicated above, and many other sections with features that network administrators use to track, monitor and secure networks.



31. Go back to PuTTY and type the command configure as highlighted above.

```
Warning: Your device is still configured with the default admin account credentials. Please change your password prior to deployment.
admin@PA-220> configure
Entering configuration mode
[edit]
```

32. Next, type in the command commit as highlighted above and the following text should be indicating how the "job" is being committed successfully.

```
admin@PA-220# commit


Commit job 2 is in progress. Use Ctrl+C to return to command prompt
...........................55%....75%.....98%................100%
Configuration committed successfully

[edit]
```

33. In a new line, type the command “set mgt-config users admin password” and type in the password you want to set for the firewall.

```
admin@PA-220# set mgt-config users admin password
Enter password   :
Confirm password :

[edit]
admin@PA-220#
```

34. As done in the previous step, type in the commit command and check if the “job” has committed successfully. Note: We commit the password again to make sure that for the few outlier cases where the password doesn’t commit, we are completely accounted for.

```
admin@PA-220# commit


Commit job 3 is in progress. Use Ctrl+C to return to command prompt
.....................
```

35. Reopen the Web GUI, look at the homepage specifically the Config Logs section as highlighted above, then make sure that the commit and set command that was typed in the previous steps match the same timestamp in order to verify if the password was committed.

Config Logs

Command	Path	Admin	Time
commit		admin	09/10 10:01:10
set	config mgt-config users admin	admin	09/10 10:00:09

36. Then in the System Logs section check if the Commit job succeeded in the Description as highlighted above match the time stamp to finally verify that you have completed the authentication commits written in a previous step.

System Logs

Description	Time
Connection to Update server closed: updates.paloaltonetworks.com, source: 192.168.1.1	09/10 10:05:06
Commit job succeeded. Completion time=2025/09/10 10:03:03. JobId=3. User:admin	09/10 10:03:03

Problems

While factory resetting the firewall, it didn't work the first time as we missed the time period for enter maintenance mode. We had to restart the firewall and try again.

The pan-OS of the firewall software is outdated. It is in Pan-os 9.2 while the most current version is 10.2. To mitigate this we need to flash the firewall with the most current image, and for that we need to set up a SOHO network.

Conclusion

The Palo Alto PA220 was previously a very unfamiliar technology to me, as I have never worked with firewalls before. However, within an hour or two, I developed a strong understanding of the PA220's console interface and learned its basic maintenance functions. I'm eager to learn what else Palo Alto firewalls have in store, and I'm excited to set these firewalls up in more advanced configurations.

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